

Code-Layout, Readability and Reusability

Date	28 September 2026
Team ID	SWTID-2026-4103
Project Name	FAQ Assistant API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are used throughout the code.	YES	Consistent indentation is used throughout the Python code.
2	Proper File Structure	Files and folders are logically organize	YES	Project files are organized logically for API development.
3	Meaningful Variable Names	Variables reflect their purpose clearly	YES	Variables such as question and answer clearly represent their purpose.
4	Function / Method Names	Functions are descriptively named	YES	Functions are named clearly according to their purpose.
5	Code Comments	Inline and block comments explain log c	YES	Comments are used where necessary to explain the code.
6	Modular Design	Code is split into reusable functions/modules	YES	The API is organized into reusable components for processing questions and responses.
7	No Redundant Code	Duplicate or unused code is removed	YES	Unnecessary and duplicate code is avoided in the project.
8	Error Handling	Exceptions and errors are handled gracefully	YES	Invalid or empty questions are handled with appropriate error responses.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	FAQ API Endpoint	Python / Fast API	Used for handling user questions	High
2	Gemini AI Integration	Python / Gemini AI	Used for generating answers	High
3	JSON Request & Response	JSON	Used for API communication	High
4	Input Validation	Python / FastAPI	Used for validating user questions	Medium
5	Error Handling	Python	Used for handling invalid requests and errors	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project has a clear and organized API structure.
Readability	5	The code is simple and easy to understand
Reusability	4	API components can be reused in other applications.
Documentation / Comments	4	Proper documentation and necessary comments are provided.
Overall Score	5	The project has good code quality, readability, and reusability.